

Unit 2 Classwork — Exponents & Scientific Notation

Exponent Rules · Power/Zero/Negative Exponents · Scientific Notation

A. Exponent Rules — Product & Quotient

Problem 1

Simplify: $x^4 \cdot x^3$

$$\underline{x^7}$$

Problem 2

Simplify: $y^8 \div y^3$

$$\underline{y^5}$$

Problem 3

Simplify: $(2x^3)(3x^2)$

$$\underline{6x^5}$$

Problem 4

Simplify: $15a^6 \div 3a^2$

$$\underline{5a^4}$$

Problem 5

Simplify: $a^2 \cdot a^5 \cdot a$

$$\underline{a^8}$$

B. Power, Zero & Negative Exponents

Problem 1

Simplify: $(x^3)^4$

$$\underline{x^{12}}$$

Problem 2

Simplify: $(2x^2)^3$

$$\underline{8x^6}$$

Problem 3

Simplify: 7^0

$$\underline{1}$$

Problem 4

Simplify: $(5x^4)^0$

1

Problem 5

Simplify: 3^{-2} (write as a fraction)

1/9

Problem 6

Simplify: x^{-4} (write as a fraction)

$1/x^4$

Problem 7

Evaluate: -4^2 vs $(-4)^2$ — what is $(-4)^2$?

16

C. Scientific Notation — Conversions**Problem 1**

Write 32,000 in scientific notation.

3.2×10^4

Problem 2

Write 0.0091 in scientific notation.

9.1×10^{-3}

Problem 3

Write 5.6×10^6 in standard form.

5,600,000

Problem 4

Write 2.3×10^{-4} in standard form.

0.00023

Problem 5

Write 470,000,000 in scientific notation.

4.7×10^8

D. Scientific Notation — Operations

Problem 1

Multiply: $(2 \times 10^3) \times (3 \times 10^5)$

$$\underline{6 \times 10^8}$$

Problem 2

Divide: $(8 \times 10^7) \div (2 \times 10^3)$

$$\underline{4 \times 10^4}$$

Problem 3

Multiply: $(6 \times 10^4) \times (5 \times 10^2)$ — adjust if needed!

$$\underline{3 \times 10^7}$$